



Practice Programs

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Program Question	
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Program 1: Write a C program to print the following text message each time it is run.
 'To C, or not to C: that is the question.'

pun.c :

```

1  #include <stdio.h>
2
3  int main(void) {
4      printf("To C, or not to C: that is the question.\n");
5      return 0;
6  }
```

Program 2: Write a C program to demonstrate simple arithmetic.

simpleArithmetic.c

```

1  #include<stdio.h>
2
3  int main(){
4      int a, b, c;
5
6      printf("Please enter two numbers : ");
7      scanf("%d %d", &a, &b); //ok
8
9      c = a + b; //addition
10     printf("The Sum is %d\n", c);
11
12     c = a - b; //subtraction
13     printf("The Difference is %d\n", c);
14
15     c = a * b; //multiplication
16     printf("The Product is %d\n", c);
17
18     c = a / b;
19     printf("The Division is %d\n", c);
20     return 0;
21 }
```

Program 3: Write a C program to calculate Simple Interest.

simpleInterestCalculator.c

```

1  #include<stdio.h>
2
3  int main(){
4      int numberOfYears;
5      float principleAmount, rateOfInterest, simpleInterest;
6
7      printf("Enter Principle Amount\n", principleAmount);
8      scanf("%f", &principleAmount);
9      printf("Enter number of year\n", numberOfYears);
10     scanf("%d", &numberOfYears);
11     printf("Enter Rate of Interest\n", rateOfInterest);
12     scanf("%f", &rateOfInterest);
13
14     simpleInterest = (principleAmount * numberOfYears * rateOfInterest) / 100;
15     printf("Simple interest is %f\n", simpleInterest);
16
17     return 0;
18 }
19

```

Program 4: Write a C program to calculate Simple EMI.

simpleEMICalculator.c

```

1  #include <stdio.h>
2  #include <math.h>
3
4  int main() {
5      float principle, rate, time, emi;
6
7      printf("Enter principle amount: ");
8      scanf("%f", &principal);
9
10     printf("\nEnter rate of interest: ");
11     scanf("%f", &rate);
12
13     printf("\nEnter time in years: ");
14     scanf("%f", &time);
15
16     rate = rate / (12 * 100); /*one month interest*/
17     time = time * 12; /*one month period*/
18
19     emi= ( principal * rate * pow(1 + rate,time)) / (pow(1 + rate,time) - 1);
20
21     printf("Monthly EMI is= %f\n", emi);
22
23     return 0;
24 }
25

```

Program 5: Write a C program to calculate Student Percentage of 5 subjects

studentPercentageCalculator.c

```

1  #include<stdio.h>
2
3  int main() {
4      int a,b,c,d,e;
5      float total, average, percentage;
6
7      printf("Enter marks of 5 subjects : \n");
8      scanf("%d %d %d %d %d", &a, &b, &c, &d, &e);
9
10     total = a + b + c + d + e;
11     average = total / 5;
12     percentage = (total / 500) * 100;
13
14     printf("Total marks = %f\n", total);
15     printf("Average marks = %f\n", average );
16     printf("Net percentage = %f\n", percentage);
17     return 0;
18 }
19

```

Program 6: Write a program taking user input of basic salary and calculating gross salary that includes basic salary, 50% DA and 40% HRA.

grossSalaryCalculator.c

```

1  #include<stdio.h>
2
3  void main() {
4      float basic,hra,da,gross;
5
6      printf("Enter the Basic Salary : $");
7      scanf("%f", &basic);
8
9      hra = 40 * basic / 100;
10     da = 50 * basic / 100;
11     gross = basic + hra + da;
12
13     printf("Gross Salary is $%f", gross);
14 }
15

```

Program 7: Write a program to calculate area of Square, Triangle & Rectangle using user inputs of shape data.

areaOfShapes.c

```

1  #include <stdio.h>
2
3  void main() {
4      float area;
5      float side; //For square
6      float base, height; //For triangle
7      float length, breadth; //For rectangle
8
9      //Area of square
10     printf("Enter the side of square in cms: ");
11     scanf("%f", &side);
12     area = side * side;
13     printf("Area of square with sides %.2f cms is %f\n", side, area);
14
15     //Area of Triangle
16     printf("Enter the base of triangle in cms : ");
17     scanf("%f", &base);
18     printf("Enter the height of triangle in cms : ");
19     scanf("%f", &height);
20     area = 0.5 * base * height;
21     printf("Area is of triangle with base %.2fcms and height %.2fcms is %fcms\n", base, height, area);
22
23     //Area of Rectangle
24     printf("Enter the length of rectangle in cms : ");
25     scanf("%f", &length);
26     printf("Enter the breadth of rectangle in cms : ");
27     scanf("%f", &breadth);
28     area = length * breadth;
29     printf("The area of rectangle with length %.2fcms and breadth %.2fcms is %fcms\n", length, breadth, area);
30 }

```


Program 8: Write a program to calculate Area & Circumference of a Circle, take user inputs.

areaAndCircumference.c

```

1  #include<stdio.h>
2
3  int main() {
4      float radius, area, circumference;
5
6      printf("Enter the radius of Circle in cms : ");
7      scanf("%f", &radius);
8
9      area = 3.14 * radius * radius; //Area of circle
10     circumference = 2 * 3.14 * radius; //Circumference of circle
11
12     printf("Area of circle is %f\n", area);
13     printf("Circumference of circle is %f\n", circumference);
14     return 0;
15 }
16

```

Program 9: Write a program to calculate Dimensional weight of a box to help shipping companies charge accordingly.

dimensionalWeight.c

```

1  /*The problem is about calculating the dimensional weight of a box,
2   which shipping companies use to charge based on space taken rather
3   than actual weight. The formula divides the box's volume by 166,
4   but since integer division in C truncates decimals (rounds down),
5   we adjust by adding 165 before dividing to properly round up.*/
6
7  #include <stdio.h>
8
9  int main(void) {
10     int height, length, width, volume, weight;
11
12     printf("Enter height of the box: ");
13     scanf("%d", &height);
14     printf("Enter length of the box: ");
15     scanf("%d", &length);
16     printf("Enter width of the box: ");
17     scanf("%d", &width);
18
19     volume = height * length * width;
20     weight = (volume + 165) / 166;
21
22     printf("Volume of box is %d (cubic inches)\n", volume);
23     printf("Dimensional weight of the box is %d pounds\n", weight);
24 }

```


Program 10: Write a program to convert the user given temperature in Celsius(*C) into Fahrenheit(*F).

celsiusToFahrenheit.c

```

1  /* Converts a Celsius temperature to Fahrenheit */
2
3  #include<stdio.h>
4
5  #define FREEZING_PT 32.0f
6  #define SCALE_FACTOR (9.0f / 5.0f)
7
8  int main() {
9      float celsius, fahrenheit;
10
11     printf("Enter the Temperature in Celcius : ");
12     scanf("%f", &celsius);
13
14     fahrenheit = SCALE_FACTOR * celsius + FREEZING_PT;
15
16     printf("Temperature in Fahernheit is %.1f F", fahrenheit);
17
18     return 0;
19 }
```

Program 11: Write a program to convert the user given temperature in Fahrenheit(*F) into Celsius(*C).

fahrenheitToCelsius.c

```

1  /* Converts a Fahrenheit temperature to Celsius */
2
3  #include <stdio.h>
4
5  #define FREEZING_PT 32.0f
6  #define SCALE_FACTOR (5.0f / 9.0f)
7
8  int main() {
9      float fahrenheit, celsius;
10
11     printf("Enter Fahrenheit temperature: ");
12     scanf("%f", &fahrenheit);
13
14     celsius = (fahrenheit - FREEZING_PT) * SCALE_FACTOR;
15
16     printf("Celsius equivalent: %.1f\n", celsius);
17
18     return 0;
19 }
```

Program 12: Write a program to convert the user given temperature in Celsius(*C) to Kelvin(*K).

celsiusToKelvin.c

```

1  /* Converts a Celsius temperature to Kelvin */
2
3  #include <stdio.h>
4
5  #define SCALE_FACTOR 273.15
6
7  int main()
8  {
9      float celsius, kelvin;
10
11     printf("Enter the Temperature in Celcius : ");
12     scanf("%f", &celsius);
13
14     kelvin = celsius + SCALE_FACTOR;
15
16     printf("Temperature in Kelvin is %.2f k\n", kelvin);
17     return 0;
18 }
```

Program 13: Write a program to convert the user given temperature in Kelvin(*K) into Celsius(*C).

kelvinToCelsius.c

```

1  /* Converts a Kelvin temperature to Celsius */
2
3  #include <stdio.h>
4
5  #define SCALE_FACTOR 273.15
6
7  int main()
8  {
9      float celsius, kelvin;
10
11     printf("Enter the Temperature in Kelvin : ");
12     scanf("%f", &kelvin);
13
14     celsius = kelvin - SCALE_FACTOR;
15
16     printf("Temperature in Celsius is %.2f C\n", celsius);
17     return 0;
18 }
```

Program 14: Write a program to accept the distance between two cities in kilometres from the user. Calculate and display this distance in meters, feet, centimetres and inches

distanceConverter.c

```

1  #include<stdio.h>
2  #include<conio.h>
3
4  void main() {
5      float km, mt, inch, ft, cm;
6
7      printf("Enter the distance between two cities in kilometers : ");
8      scanf("%f", &km);
9
10     mt = km * 1000;
11     ft = mt * 3.33;
12     cm = mt * 100;
13     inch = ft * 12;
14
15     printf("The distance in meters is = %.2f mts.\n", mt);
16     printf("The distance in feets is = %.2f ft.\n", ft);
17     printf("The distance in centimeters is = %.2f cms.\n", cm);
18     printf("The distance in inchs is = %.2f inches.\n", inch);
19 }
```

Program 15: Write a program to demonstrate bitwise operator on two integers.

bitwiseCalculation.c

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int a = 14, b = 7;
6
7      printf("Bitwise NOT of a (~a): %d\n", ~a);
8      printf("Bitwise NOT of b (~b): %d\n\n", ~b);
9
10     printf("a & b = %d\n", a & b);
11
12     printf("a | b = %d\n", a | b);
13
14     printf("a ^ b = %d\n\n", a ^ b);
15
16     printf("a << 1 = %d\n", a << 1);
17     printf("b << 1 = %d\n\n", b << 1);
18
19     printf("a >> 1 = %d\n", a >> 1);
20     printf("b >> 1 = %d\n", b >> 1);
21
22     return 0;
23 }
```

Program 16: Write a program to demonstrate increment operator on an integer.

incrementOperator.c

```

1 // Increment operator
2
3 #include<stdio.h>
4 int main()
5 {
6     int a= 5;
7     int b, c, d;
8
9     printf("Value of a = %d\n", a);
10
11     b = ++a;
12     c = a++;
13     d = ++a;
14
15     printf("Value of b using ++a = %d\n", b);
16     printf("Value of c using a++ = %d\n", c);
17     printf("Value of d using ++a = %d\n", d);
18
19     return 0;
20 }
```

Program 17: Write a program to demonstrate decrement operator on an integer.

decrementOperator.c

```

1 // Decrement operator
2
3 #include<stdio.h>
4
5 int main() {
6     int a= 8;
7     int b, c, d;
8
9     printf("Value of a = %d\n", a);
10
11     b = --a;
12     c = a--;
13     d = --a;
14
15     printf("Value of b using --a = %d\n", b);
16     printf("Value of c using a-- = %d\n", c);
17     printf("Value of d using --a = %d\n", d);
18
19     return 0;
20 }
```

Program 18: Write a program to calculate size of variables.

calculateSizeOfOperator.c

```

1  /* sizeof() operator in c is used to calculate the size of a variable
2  used inside the program and return the size in integer in the form
3  of memory bytes*/
4
5  #include<stdio.h>
6
7  int main()
8  {
9      int a = 6;
10     float b = 8.765f;
11     long long int c = 15050603LL;
12     double d = 78945.321654;
13     char ch = 'X';    // Capital X
14
15     printf("Size of a : %d Bytes\n" ,sizeof(a));
16     printf("Size of b : %d Bytes\n" ,sizeof(b));
17     printf("Size of c : %d Bytes\n" ,sizeof(c));
18     printf("Size of d : %d Bytes\n" ,sizeof(d));
19     printf("Size of X : %d Bytes\n",sizeof(ch));
20
21     return 0;
22 }
```

Program 19: Write a program to find ASCII value of the character user inputs.

findASCIIValue.c

```

1  /* Program to find ASCII value */
2
3  #include <stdio.h>
4
5  int main() {
6      char c;
7
8      printf("Enter a character: ");
9
10     // Reads character input from the user
11     scanf("%c", &c);
12
13     // %d displays the integer value of a character
14     // %c displays the actual character
15     printf("ASCII value of %c = %d", c, c);
16     return 0;
17 }
```


Program 20: Write a program to output numbers in certain formats using printf() function.

formattedOutput.c

```

1  /* Prints int and float values in various formats */
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      int i;
8      float x;
9
10     i = 40;
11     x = 839.21f;
12
13     printf("|%d|%5d|%-5d|%5.3d|\n", i, i, i, i);
14     printf("|%10.3f|%10.3e|%-10g|\n", x, x, x);
15
16     return 0;
17 }
```

Program 21: Write a program to add two fractions taken from the user and show the result.

addTwoFraction;

```

1  /* Adds two fractions */
2
3  #include <stdio.h>
4
5  int main(void) {
6      int num1, denom1, num2, denom2, result_num, result_denom;
7
8      printf("Enter first fraction: ");
9      scanf("%d/%d", &num1, &denom1);
10
11     printf("Enter second fraction: ");
12     scanf("%d/%d", &num2, &denom2);
13
14     result_num = num1 * denom2 + num2 * denom1;
15     result_denom = denom1 * denom2;
16
17     printf("The sum is %d/%d\n", result_num, result_denom);
18
19     return 0;
20 }
```

Program 22: Write a program to add two fractions taken from the user and show the result.

upcCodeCheck.c

```

1  /* Computes a Universal Product Code check digit */
2
3  #include <stdio.h>
4
5  int main(void) {
6      int d, i1, i2, i3, i4, i5, j1, j2, j3, j4, j5;
7      int first_sum, second_sum, total;
8
9      printf("Enter the first (single) digit: ");
10     scanf("%d", &d);
11     printf("Enter first group of five digits: ");
12     scanf("%d%d%d%d%d", &i1, &i2, &i3, &i4, &i5);
13     printf("Enter second group of five digits: ");
14     scanf("%d%d%d%d%d", &j1, &j2, &j3, &j4, &j5);
15
16     first_sum = d + i2 + i4 + j1 + j3 + j5;
17     second_sum = i1 + i3 + i5 + j2 + j4;
18     total = 3 * first_sum + second_sum;
19
20     printf("Check digit: %d\n", 9 - ((total - 1) % 10));
21
22     return 0;
23 }
```

Program 23: Write a program to display the marksheet of the students.

studentMarksheet.c

```

1  /*Evaluate total, average and percentage of a student*/
2
3  #include<stdio.h>
4
5  int main() {
6      int a,b,c,d,e;
7      float total,average,percentage;
8
9      printf("Enter marks of 5 subjects : \n");
10     scanf("%d %d %d %d %d",&a,&b,&c,&d,&e);
11
12     total = a+b+c+d+e;
13     average = total/5;
14     percentage = (total/500)*100;
15
16     printf("Total marks = %f\n", total);
17     printf("Average marks = %f\n",average );
18     printf("Net percentage = %f\n",percentage);
19     return 0;
20 }
```


Program 24: Write a program to calculate tan angle of a given angle.

tanAngle.c

```

1  /*Calculates tan angle of the given angle*/
2
3  #include <stdio.h>
4  #include <math.h>
5
6  void main () {
7      double a, b;
8
9      printf ("Enter the degree: ");
10
11     scanf ("%lf", &a);
12
13     a = (a * 3.14) / 180;
14     b = tan(a);
15
16     printf("tan is %lf", b);
17 }
```

Program 25: Write a program to calculate log value of a given number.

logarithm.c

```

1  /*Program to calculate log */
2
3  #include <stdio.h>
4  #include <math.h>
5
6  int main(int argc, const char * argv[]) {
7      /* Define temporary variables */
8      double value, result;
9
10     printf("Enter the value of Log to calculate: ");
11     scanf("%lf", &value);
12
13     /* Calculate the log of the value */
14     result = log(value);
15
16     /* Display the result of the calculation */
17     printf("The Natural Logarithm of %lf is %lf\n", value, result);
18
19     return 0;
20 }
```

Program 26: Write a program to check the type of input character given by user.

checkCharacterInput.c

```

1 //program to check type of input character.
2
3 #include<stdio.h>
4
5 int main() {
6     char ch;
7
8     printf("Enter a character: ");
9     scanf("%c",&ch);
10
11     if((ch >= 65 && ch <= 90) || (ch >= 97 && ch <= 122)){
12         printf("Entered character is an alphabet");
13     }else if (ch >= 48 && ch <= 57) {
14         printf("Entered character is a digit");
15     }else {
16         printf("Entered character is a special symbol");
17     }
18     return 0;
19 }
```

Program 27: Write a program to check if the year given by user is a leap year or not.

isLeapYear.c

```

1 /*Program checks if the given year is a leap year or not*/
2 #include <stdio.h>
3
4 int main() {
5     int input,i;
6
7     printf("Please input a year and I will tell you if it's a leap year: ");
8     scanf("%d",&input);
9
10    if((input % 4 == 0 && input % 100 != 0) || input % 400 == 0) {
11        printf("Your input (%d) IS a leap year.\n\n",input);
12    } else {
13        printf("Your input (%d) is NOT a leap year.\n\n",input);
14    }
15
16    return 0;
17 }
```

Program 28: Write a program to check if the year given by user is a leap year or not using ternary operator.

isLeapYearTernary.c

```

1  /*Program to check for leap year using ternary operator*/
2
3  #include<stdio.h>
4
5  int main() {
6      int y;
7
8      printf("Please input a year and I will tell you if it's a leap year: ");
9      scanf("%d",&y);
10
11     //Using ternary operator
12     (((y % 100 != 0) && (y % 4 == 0)) || (y % 400 == 0)) ?
13         printf("Your input (%d) is a leap year.",y) :
14         printf("Your input (%d) is NOT a leap year.",y);
15
16     return 0;
17 }

```

Program 29: Write a program to check the greatest number given by user.

greatestNumber.c

```

1  /*Greatest number using nested if statement*/
2
3  #include<stdio.h>
4
5  int main()
6  {
7      int a,b,c;
8      printf("Enter three integer number: ");
9      scanf("%d %d %d",&a,&b,&c);
10
11     if(a>b) {
12         if(a>c)
13             printf("%d is greatest", a);
14         else
15             printf("%d is greatest", c);
16     } else {
17         if(b>c)
18             printf("%d is greatest", b);
19         else
20             printf("%d is greatest", c);
21     }
22     return 0;
23 }

```

Program 30: Write a program to calculate the commission of broker on a trade value.

brokersCommission.c

```

1  /* Calculates a broker's commission */
2
3  #include <stdio.h>
4
5  int main(void) {
6      float commission, value;
7
8      printf("Enter value of trade: ");
9      scanf("%f", &value);
10
11     if (value < 2500.00f) {
12         commission = 30.00f + .017f * value;
13     } else if (value < 6250.00f) {
14         commission = 56.00f + .0066f * value;
15     } else if (value < 20000.00f) {
16         commission = 76.00f + .0034f * value;
17     } else if (value < 50000.00f) {
18         commission = 100.00f + .0022f * value;
19     } else if (value < 500000.00f) {
20         commission = 155.00f + .0011f * value;
21     } else {
22         commission = 255.00f + .0009f * value;
23     }
24
25     if (commission < 39.00f) {
26         commission = 39.00f;
27     }
28
29     printf("Commission: $%.2f\n", commission);
30
31     return 0;
32 }
```

Program 31: Write a program to output a digit given by user in words.

digitToWords.c

```

1  /*Program to output digits in words*/
2
3  #include <stdio.h>
4
5  int main() {
6      int digit;
7
8      printf("Enter a single digit number (0-9): ");
9      scanf("%d", &digit);
10
11     switch (digit) {
12         case 0:
13             printf("Zero\n");
14             break;
15         case 1:
16             printf("One\n");
17             break;
18         case 2:
19             printf("Two\n");
20             break;
21         case 3:
22             printf("Three\n");
23             break;
24         case 4:
25             printf("Four\n");
26             break;
27         case 5:
28             printf("Five\n");
29             break;
30         case 6:
31             printf("Six\n");
32             break;
33         case 7:
34             printf("Seven\n");
35             break;
36         case 8:
37             printf("Eight\n");
38             break;
39         case 9:
40             printf("Nine\n");
41             break;
42         default:
43             printf("Invalid input! Please enter a single digit (0-9).\n");
44     }
45
46     return 0;
47 }

```


Program 32: Write a program to give simple calculator options to user.

simpleCalculator.c

```

1  /*Program for simple calculator*/
2
3  #include <stdio.h>
4
5  int main() {
6      int choice;
7      float num1, num2, result;
8
9      printf("==== Simple Calculator ==== \n");
10     printf("1. Addition \n");
11     printf("2. Subtraction \n");
12     printf("3. Multiplication \n");
13     printf("4. Division \n");
14     printf("5. Modulus \n");
15     printf("Enter your choice (1-5): ");
16     scanf("%d", &choice);
17
18     // For modulus, we'll use int inputs, so read accordingly
19     if (choice >= 1 && choice <= 5) {
20         printf("Enter two numbers: ");
21         scanf("%f %f", &num1, &num2);
22     }
23
24     switch (choice) {
25         case 1:
26             result = num1 + num2;
27             printf("Result = %.2f \n", result);
28             break;
29
30         case 2:
31             result = num1 - num2;
32             printf("Result = %.2f \n", result);
33             break;
34
35         case 3:
36             result = num1 * num2;
37             printf("Result = %.2f \n", result);
38             break;
39
40         case 4:
41             if (num2 != 0)
42                 printf("Result = %.2f \n", num1 / num2);
43             else
44                 printf("Error! Division by zero not allowed. \n");
45             break;
46
47         case 5:
48             // Modulus only makes sense for integers
49             printf("Result = %d \n", (int)num1 % (int)num2);
50             break;
51
52         default:
53             printf("Invalid choice! Please enter between 1 and 5. \n");
54     }
55
56     return 0;
57 }

```


Program 33: Write a program to output date in legal form.

date.c

```

1  /* Prints a date in legal form */
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      int month, day, year;
8
9      printf("Enter date (mm/dd/yy): ");
10     scanf("%d /%d /%d", &month, &day, &year);
11
12     printf("Dated this %d", day);
13     switch (day) {
14         case 1: case 21: case 31:
15             printf("st"); break;
16         case 2: case 22:
17             printf("nd"); break;
18         case 3: case 23:
19             printf("rd"); break;
20         default: printf("th"); break;
21     }
22     printf(" day of ");
23
24     switch (month) {
25         case 1: printf("January"); break;
26         case 2: printf("February"); break;
27         case 3: printf("March"); break;
28         case 4: printf("April"); break;
29         case 5: printf("May"); break;
30         case 6: printf("June"); break;
31         case 7: printf("July"); break;
32         case 8: printf("August"); break;
33         case 9: printf("September"); break;
34         case 10: printf("October"); break;
35         case 11: printf("November"); break;
36         case 12: printf("December"); break;
37     }
38
39     printf(", 20%.2d.\n", year);
40     return 0;
41 }

```

Program 34: Write a program check if the number is automorphic or not.

automorphic.c

```

1  /*Program to show if the number is Automorphic*/
2
3  #include<stdio.h>
4  #include<stdlib.h>
5  #include<math.h>
6
7  int main() {
8      int num, sqr, temp, last, n = 0;
9
10     printf("Enter a number \n");
11     scanf("%d", &num);
12     sqr = num * num; //calculating square of num
13     temp = num;
14
15     //Counting number of digits
16     while (temp > 0) {
17         n++;
18         temp = temp / 10;
19     }
20
21     //Extracting last n digits
22     int den = floor(pow(10, n));
23     last = sqr % den;
24
25     if (last == num)
26         printf("Automorphic number \n");
27     else
28         printf("Not Automorphic \n");
29
30     return 0;
31 }
```

Program 35: Write a program show linear search.

linearSearch.c

```
1  /*Program to show linear search*/
2
3  #include<stdio.h>
4
5  int main() {
6      int n, i, x, c = 0;
7
8      printf("ENTER SIZE OF ARRAY: ");
9      scanf("%d", &n);
10
11     int a[n];
12
13     for(i=0;i<n;i++) {
14         scanf("%d", &a[i]);
15     }
16
17     printf("ENTER ELEMENT TO SEARCH\n");
18     scanf("%d",&x);
19
20     for(i=0;i<n;i++) {
21         if(a[i]==x) {
22             printf("FOUND AT INDEX: %d i.e. ELEMENT NUMBER: %d", i, i+1);
23             c++;
24             break;
25         }
26     }
27
28     if(c==0) {
29         printf("ELEMENT NOT FOUND");
30     }
31
32     return 1;
33 }
```

Program 36: Write a program to check if the given number is perfect number.

perfectNumber.c

```

1  /*Program to check if the number is perfect or not*/
2
3  #include<stdio.h>
4
5  int main () {
6      int n, i, sum = 0;
7
8      printf("Enter the value of n: ");
9      scanf("%d", &n);
10
11     for (i = 1; i < n; i++) {
12         if (n % i == 0)
13             sum = sum + i;
14     }
15
16     if (n == sum)
17         printf("Entered no. is a perfect no.");
18     else
19         printf("Entered no. is not a perfect no.");
20
21     return 0;
22 }
```

Program 37: Write a program to check if the number is prime number.

primeNumber.c

```

1  /*Program to check if the number is prime*/
2
3  #include<stdio.h>
4
5  void main() {
6      int i, n, c = 0;
7
8      printf("Enter a Number : ");
9      scanf("%d", &n);
10
11     for(i = 2; i < n; i++){
12         if(n % i == 0) {
13             printf("NOT a prime number");
14             c++;
15             break;
16         }
17     }
18
19     if(c == 0)
20         printf("%d is a prime number", n);
21 }
```

Program 38: Write a program to print simple multiplication table of a given number.

simpleMultiplicationTable.c

```

1  /*Program to display multiplication table*/
2
3  #include <stdio.h>
4
5  int main() {
6      int num, i = 1;
7
8      printf("Enter a number to calculate the multiplication table up to 10: \n");
9      scanf("%d", &num);
10
11     for(i = 1; i <= 10; i++) {
12         printf("%d x %d = %d \n", i, num, num * i);
13     }
14
15     return 0;
16 }

```

Program 39: Write a program to reverse a number.

reverseNumber.c

```

1  /*Program to Reverse a number*/
2
3  #include <stdio.h>
4
5  int main() {
6      int num, reverse = 0, remainder;
7
8      printf("Enter a number\n");
9      scanf("%d", & num);
10
11     while (num != 0) {
12         remainder = num % 10;
13         reverse = reverse * 10 + remainder;
14         num = num / 10;
15     }
16
17     printf("Reverse of the number is : %d\n", reverse);
18     return 0;
19 }

```

Program 40: Write a program to reverse a given string.

stringReverse.c

```
1  /*Program to reverse a string*/
2
3  #include<stdio.h>
4  #include<conio.h>
5
6  void main() {
7      int i, j, k;
8      char str[100];
9      char rev[100];
10
11     printf("Enter a string:\t");
12     scanf("%s", str);
13     printf("The original string is %s\n", str);
14
15     for(i = 0; str[i] != '\0'; i++);
16
17     k = i-1;
18
19     for(j = 0; j <= i-1; j++){
20         rev[j] = str[k];
21         k--;
22     }
23
24     printf("The reverse string is %s\n", rev);
25     getch();
26 }
```


Program 41: Write a program to output * pattern.

starPattern.c

```

1  /*Program to print pattern*/
2
3  #include <stdio.h>
4
5  int main(){
6      int n, i, j;
7
8      printf("Enter number of rows : ");
9      scanf("%d", &n);
10
11     for(i = 1; i <= n; i++){
12         for(j = 1; j <= i; j++) {
13             printf("* ");
14         }
15         printf("\n");
16     }
17
18     return 0;
19 }
```

Program 42: Write a program to output * pattern.

starPattern2.c

```

1  /*Program to print pattern*/
2
3  #include <stdio.h>
4
5  int main() {
6      int n, i, j, space;
7      printf("Enter number of rows: ");
8      scanf("%d", &n);
9
10     for(i = 1; i <= n; i++) {
11         for(space = 1; space <= n - i; space++) {
12             printf(" ");
13         }
14         for(j = 1; j <= (2 * i - 1); j++) {
15             printf("*");
16         }
17         printf("\n");
18     }
19
20     return 0;
21 }
```

Program 43: Write a program to output number pattern.

numberPattern.c

```

1  /*Number pattern program*/
2
3  #include <stdio.h>
4
5  int main() {
6      int i, j, rows, num = 1;
7
8      printf("Enter number of rows: ");
9      scanf("%d", &rows);
10
11     for(i = 1; i <= rows; i++) {
12         for(j = 1; j <= i; j++) {
13             printf("%d ", num);
14             num++;
15         }
16         printf("\n");
17     }
18
19     return 0;
20 }
```

Program 44: Write a program to output palindromic number pattern.

numberPattern2.c

```

1  /*Number palindromic pattern program*/
2
3  #include <stdio.h>
4
5  int main() {
6      int n, i, j, k;
7      printf("Enter number of rows: ");
8      scanf("%d", &n);
9
10     for(i = 1; i <= n; i++) {
11         for(j = 1; j <= n - i; j++) {
12             printf(" ");
13         }
14         for(k = i; k >= 1; k--) {
15             printf("%d ", k);
16         }
17         for(k = 2; k <= i; k++) {
18             printf("%d ", k);
19         }
20         printf("\n");
21     }
22
23     return 0;
24 }
```

Program 45: Write a program to output Alphabet Diamond Pattern.

characterPattern.c

```

1  /*Character diamond pattern program*/
2
3  #include <stdio.h>
4
5  int main() {
6      int n, i, j, space;
7      printf("Enter number of rows (e.g., 5): ");
8      scanf("%d", &n);
9
10     // Upper half
11     for(i = 1; i <= n; i++) {
12         for(space = 1; space <= n - i; space++) {
13             printf(" ");
14         }
15         char ch = 'A';
16         for(j = 1; j <= i; j++) {
17             printf("%c", ch);
18             ch++;
19         }
20         ch -= 2;
21         for(j = 1; j < i; j++) {
22             printf("%c", ch);
23             ch--;
24         }
25         printf("\n");
26     }
27
28     // Lower half
29     for(i = n - 1; i >= 1; i--) {
30         for(space = 1; space <= n - i; space++) {
31             printf(" ");
32         }
33         char ch = 'A';
34         for(j = 1; j <= i; j++) {
35             printf("%c", ch);
36             ch++;
37         }
38         ch -= 2;
39         for(j = 1; j < i; j++) {
40             printf("%c", ch);
41             ch--;
42         }
43         printf("\n");
44     }
45
46     return 0;
47 }

```

Program 46: Write a program to output Alphabet Pyramid.

characterPattern2.c

```

1  /*Character pyramid pattern program*/
2
3  #include <stdio.h>
4
5  int main() {
6      int n, i, j, space;
7      printf("Enter number of rows: ");
8      scanf("%d", &n);
9
10     for(i = 1; i <= n; i++) {
11         for(space = 1; space <= n - i; space++) {
12             printf(" ");
13         }
14         char ch = 'A';
15         for(j = 1; j <= i; j++) {
16             printf("%c ", ch);
17             ch++;
18         }
19         printf("\n");
20     }
21
22     return 0;
23 }
```

Program 47: Write a program to output Alphabet–Number Center Pyramid.

characterNumberPattern.c

```

1  /*Character number pyramid pattern program*/
2
3  #include <stdio.h>
4
5  int main() {
6      int n, i, j, space;
7      printf("Enter number of rows: ");
8      scanf("%d", &n);
9
10     for(i = 1; i <= n; i++) {
11         for(space = 1; space <= n - i; space++) {
12             printf(" ");
13         }
14         char ch = 'A';
15         for(j = 1; j <= i; j++) {
16             printf("%c ", ch++);
17         }
18         int num = 1;
19         for(j = 1; j <= i; j++) {
20             printf("%d ", num++);
21         }
22         printf("\n");
23     }
24
25     return 0;
26 }
```

Program 48: Write a program to calculate the sum of given integers.

sum.c

```

1  /* Sums a series of numbers */
2
3  #include <stdio.h>
4
5  int main(void) {
6      int n, sum = 0;
7
8      printf("This program sums a series of integers.\n");
9      printf("Enter integers (0 to terminate): ");
10
11     scanf("%d", &n);
12     while (n != 0) {
13         sum += n;
14         scanf("%d", &n);
15     }
16     printf("The sum is: %d\n", sum);
17
18     return 0;
19 }
```

Program 49: Write a program to calculate the length of the message.

lengthOfMessage.c

```

1  /* Determines the length of a message */
2
3  #include <stdio.h>
4
5  int main(void) {
6      char ch;
7      int len = 0;
8
9      printf("Enter a message: ");
10     ch = getchar();
11
12     while (ch != '\n') {
13         len++;
14         ch = getchar();
15     }
16     printf("Your message was %d character(s) long.\n", len);
17
18     return 0;
19 }
```

Program 50: Write a program demonstrate a real-life check book using switch statement.

checkbook.c

```

1  /* Balances a checkbook */
2
3  #include <stdio.h>
4
5  int main(void) {
6      int cmd;
7      float balance = 0.0f, credit, debit;
8
9      printf("*** ACME checkbook-balancing program ***\n");
10     printf("Commands: 0=clear, 1=credit, 2=debit, ");
11     printf("3=balance, 4=exit\n\n");
12
13     for (;;) {
14         printf("Enter command: ");
15         scanf("%d", &cmd);
16         switch (cmd) {
17             case 0:
18                 balance = 0.0f;
19                 break;
20             case 1:
21                 printf("Enter amount of credit: ");
22                 scanf("%f", &credit);
23                 balance += credit;
24                 break;
25             case 2:
26                 printf("Enter amount of debit: ");
27                 scanf("%f", &debit);
28                 balance -= debit;
29                 break;
30             case 3:
31                 printf("Current balance: $%.2f\n", balance);
32                 break;
33             case 4:
34                 return 0;
35             default:
36                 printf("Commands: 0=clear, 1=credit, 2=debit, ");
37                 printf("3=balance, 4=exit\n\n");
38                 break;
39         }
40     }
41 }
```


Program 51: Write a program to reverse a series a number.

reverseSeriesOfNumber.c

```

1  /* Reverses a series of numbers */
2
3  #include <stdio.h>
4  #define N 10
5
6  int main(void) {
7      int a[N], i;
8
9      printf("Enter %d numbers: ", N);
10     for (i = 0; i < N; i++) {
11         scanf("%d", &a[i]);
12     }
13
14     printf("In reverse order:");
15     for (i = N - 1; i >= 0; i--) {
16         printf(" %d", a[i]);
17     }
18     printf("\n");
19
20     return 0;
21 }

```

Program 52: Write a program to check if the number consists of a repeated digit.

repeatedDigits.c

```

1  /* Checks numbers for repeated digits */
2
3  #include <stdbool.h>
4  #include <stdio.h>
5
6  int main(void) {
7      bool digit_seen[10] = {false};
8      int digit;
9      long n;
10
11     printf("Enter a number: ");
12     scanf("%ld", &n);
13
14     while (n > 0) {
15         digit = n % 10;
16         if (digit_seen[digit])
17             break;
18         digit_seen[digit] = true;
19         n /= 10;
20     }
21
22     if (n > 0)
23         printf("Repeated digit\n");
24     else
25         printf("No repeated digit\n");
26
27     return 0;
28 }

```

Program 53: Write a program to output a number a random card.

cardsRandom.c

```

1  /* Deals a random hand of cards */
2
3  #include <stdbool.h>
4  #include <stdio.h>
5  #include <stdlib.h>
6  #include <time.h>
7
8  #define NUM_SUITS 4
9  #define NUM_RANKS 13
10
11 int main(void) {
12     bool in_hand[NUM_SUITS][NUM_RANKS] = {false};
13     int num_cards, rank, suit;
14     const char rank_code[] = {'2','3','4','5','6','7','8',
15                               '9','t','j','q','k','a'};
16     const char suit_code[] = {'c','d','h','s'};
17
18     srand((unsigned) time(NULL));
19
20     printf("Enter number of cards in hand: ");
21     scanf("%d", &num_cards);
22
23     printf("Your hand:");
24     while (num_cards > 0) {
25         suit = rand() % NUM_SUITS;      /* picks a random suit */
26         rank = rand() % NUM_RANKS;      /* picks a random rank */
27         if (!in_hand[suit][rank]) {
28             in_hand[suit][rank] = true;
29             num_cards--;
30             printf(" %c%c", rank_code[rank], suit_code[suit]);
31         }
32     }
33     printf("\n");
34
35     return 0;
36 }

```

Program 54: Write a program to print the table of interest according to the years.

interestTable.c

```

1  /* Prints a table of compound interest */
2
3  #include <stdio.h>
4  #define NUM_RATES ((int) (sizeof(value) / sizeof(value[0])))
5  #define INITIAL_BALANCE 100.00
6
7  int main(void) {
8      int i, low_rate, num_years, year;
9      double value[5];
10
11     printf("Enter interest rate: ");
12     scanf("%d", &low_rate);
13     printf("Enter number of years: ");
14     scanf("%d", &num_years);
15
16     printf("\nYears");
17     for (i = 0; i < NUM_RATES; i++) {
18         printf("%6d%%", low_rate + i);
19         value[i] = INITIAL_BALANCE;
20     }
21     printf("\n");
22
23     for (year = 1; year <= num_years; year++) {
24         printf("%3d", year);
25         for (i = 0; i < NUM_RATES; i++) {
26             value[i] += (low_rate + i) / 100.0 * value[i];
27             printf("%7.2f", value[i]);
28         }
29         printf("\n");
30     }
31
32     return 0;
33 }

```

Program 55: Write a program to print pun using function.

punFunction.c

```

1  /* Prints a bad pun */
2
3  #include <stdio.h>
4
5  void print_pun(void) {
6     printf("To C, or not to C: that is the question.\n");
7 }
8
9  int main(void) {
10     print_pun();
11     return 0;
12 }

```

Program 56: Write a program to give average of the given number.

average.c

```

1  /* Computes pairwise averages of three numbers */
2
3  #include <stdio.h>
4
5  double average(double a, double b) {
6      return (a + b) / 2;
7  }
8
9  int main(void) {
10     double x, y, z;
11
12     printf("Enter three numbers: ");
13     scanf("%lf%lf%lf", &x, &y, &z);
14     printf("Average of %g and %g: %g\n", x, y, average(x, y));
15     printf("Average of %g and %g: %g\n", y, z, average(y, z));
16     printf("Average of %g and %g: %g\n", x, z, average(x, z));
17
18     return 0;
19 }

```

Program 57: Write a program to give countdown of the user given number.

countDown.c

```

1  /* Prints a countdown */
2
3  #include <stdio.h>
4
5  void printCount(int n) {
6      printf("T minus %d and counting\n", n);
7  }
8
9  int main(void) {
10     int i, n;
11     printf("Enter the number to start countdown from: ");
12     scanf("%d", &n);
13
14     for (i = n; i > 0; --i) {
15         printCount(i);
16     }
17
18     return 0;
19 }

```

Program 58: Write a program check if the number is prime using a function.

primeFunction.c

```

1  /* Tests whether a number is prime */
2
3  #include <stdbool.h>
4  #include <stdio.h>
5
6  bool isPrime(int n) {
7      int divisor;
8
9      if (n <= 1)
10         return false;
11
12     for (divisor = 2; divisor * divisor <= n; divisor++) {
13         if (n % divisor == 0)
14             return false;
15     }
16
17     return true;
18 }
19
20 int main(void) {
21     int n;
22
23     printf("Enter a number: ");
24     scanf("%d", &n);
25     if (isPrime(n))
26         printf("Prime\n");
27     else
28         printf("Not prime\n");
29
30     return 0;
31 }

```

Program 59: Write a program to give factorial of a given number.

factorial.c

```

1  /*Program to output Factorial number*/
2
3  #include <stdio.h>
4
5  // Function to calculate factorial
6  int factorial(int n) {
7      int fact = 1;
8
9      for(int i = 1; i <= n; i++) {
10         fact *= i; // multiply fact by each number up to n
11     }
12
13     return fact;
14 }
15
16 int main() {
17     int num;
18
19     printf("Enter a number: ");
20     scanf("%d", &num);
21
22     printf("Factorial of %d = %d", num, factorial(num));
23     return 0;
24 }

```

Program 60: Write a program to check if the number is Armstrong number.

armstrongNumber.c

```

1  /* Function to check Armstrong number*/
2
3  #include <stdio.h>
4  #include <math.h>
5
6  int isArmstrong(int n) {
7      int temp = n, sum = 0, digits = 0;
8
9      // Count digits
10     while(temp != 0) {
11         digits++;
12         temp /= 10;
13     }
14     temp = n;
15     while(temp != 0) {
16         int r = temp % 10;
17         sum += pow(r, digits); // r^digits
18         temp /= 10;
19     }
20
21     if(sum == n)
22         return 1; // Armstrong
23     else
24         return 0; // Not Armstrong
25 }
26
27 int main() {
28     int num;
29     printf("Enter a number: ");
30     scanf("%d", &num);
31
32     if(isArmstrong(num))
33         printf("%d is an Armstrong number", num);
34     else
35         printf("%d is NOT an Armstrong number", num);
36
37     return 0;
38 }

```


Program 61: Write a program to give Fibonacci series.

fibonnaciSeries.c

```
1  /*Program to print fibonnaci series*/
2
3  #include <stdio.h>
4
5  // Function to print Fibonacci series
6  void fibonacci(int n) {
7      int a = 0, b = 1, c;
8      printf("Fibonacci Series: %d %d ", a, b);
9
10     for(int i = 3; i <= n; i++) {
11         c = a + b;
12         printf("%d ", c);
13         a = b;
14         b = c;
15     }
16 }
17
18 int main() {
19     int n;
20     printf("Enter number of terms: ");
21     scanf("%d", &n);
22
23     fibonacci(n);
24     return 0;
25 }
```

Program 62: Write a program to demonstrate quick sort algorithm.

quickSort.c

```

1  /* Sorts an array of integers using Quicksort algorithm */
2
3  #include <stdio.h>
4
5  #define N 10
6
7  void quicksort(int a[], int low, int high);
8  int split(int a[], int low, int high);
9
10 int main(void) {
11     int a[N], i;
12
13     printf("Enter %d numbers to be sorted: ", N);
14     for (i = 0; i < N; i++) {
15         scanf("%d", &a[i]);
16     }
17
18     quicksort(a, 0, N - 1);
19
20     printf("In sorted order: ");
21     for (i = 0; i < N; i++) {
22         printf("%d ", a[i]);
23     }
24     printf("\n");
25
26     return 0;
27 }
28
29 void quicksort(int a[], int low, int high) {
30     int middle;
31
32     if (low >= high) return;
33     middle = split(a, low, high);
34     quicksort(a, low, middle - 1);
35     quicksort(a, middle + 1, high);
36 }
37
38 int split(int a[], int low, int high) {
39     int part_element = a[low];
40
41     for (;;) {
42         while (low < high && part_element <= a[high]) {
43             high--;
44         }
45         if (low >= high) break;
46
47         a[low++] = a[high];
48
49         while (low < high && a[low] <= part_element) {
50             low++;
51         }
52         if (low >= high) break;
53
54         a[high--] = a[low];
55     }
56
57     a[high] = part_element;
58     return high;
59 }

```